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Take home Quiz 1.

1. Lists

a. Two real-world scenarios where lists may be used

1. In a todo list where saving tasks to be done
2. In a company database, saving the names of the workers of the company.

b. why other structures aren't suitable ?

Sets: They dont support duplicate entries, yet in a to-do list, a user may need to do repeated tasks

Dictionary: We don't need keys to save worker's names.

Tuples: Tuples are immutable, yet data like names or tasks may need to be updated.

C. Usage/structure:

Workers List = ['peter', 'paul', 'Blaise']

2. Sets

a. Two real-world scenarios where sets may be used:

1. A Government system that saves national Ids
2. A system that holds unique phones numbers that called the call center.

b. why other data structures aren't suitable?

Lists: Data may be duplicated (phone numbers or national Id numbers)

Dictionaries: The interest is already only on unique values. we don't need keys.

Tuples: we can't be able to add more data on tuples.

C. Usage / structure

phone_numbers = {"+230790076067", "+2107818 1415"}

3. Dictionary

a. Two real-world scenarios where Dictionary is used.

- In a sales management system when saving product price information
- A school ~~ma~~ system ~~nay~~ may use dictionary in saving the students details.

b. why other data structures are not suitable.

Lists: don't support key-value relationships, seaching in bigger data may be harder.

sets: sets only hold values; they can't support associating keys with values.

Tuples: they don't support key-value pairing because they are immutable

c. Usage/structure

prices = {"water": 3, "kalypoo": 5.8, "bread": 5}

4. Tuples

a Two real-world scenarios where Tuples is used

1. the system that stores the relationships between students and their room numbers
2. a system that holds GPS Coordinates

b. why other data structures are not suitable.

Lists: Allow changes yet Foreign keys can never change

set: set can't save duplicated values.

Dictionary: may add unnecessary complexity yet we are only interested

C. usage/structure:

assignment = ("Student473", "Room 12")